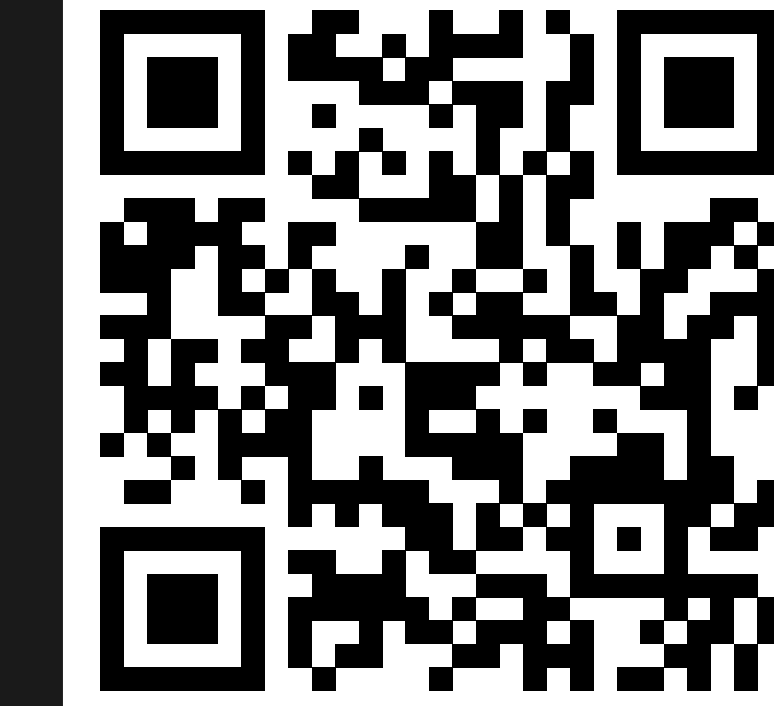




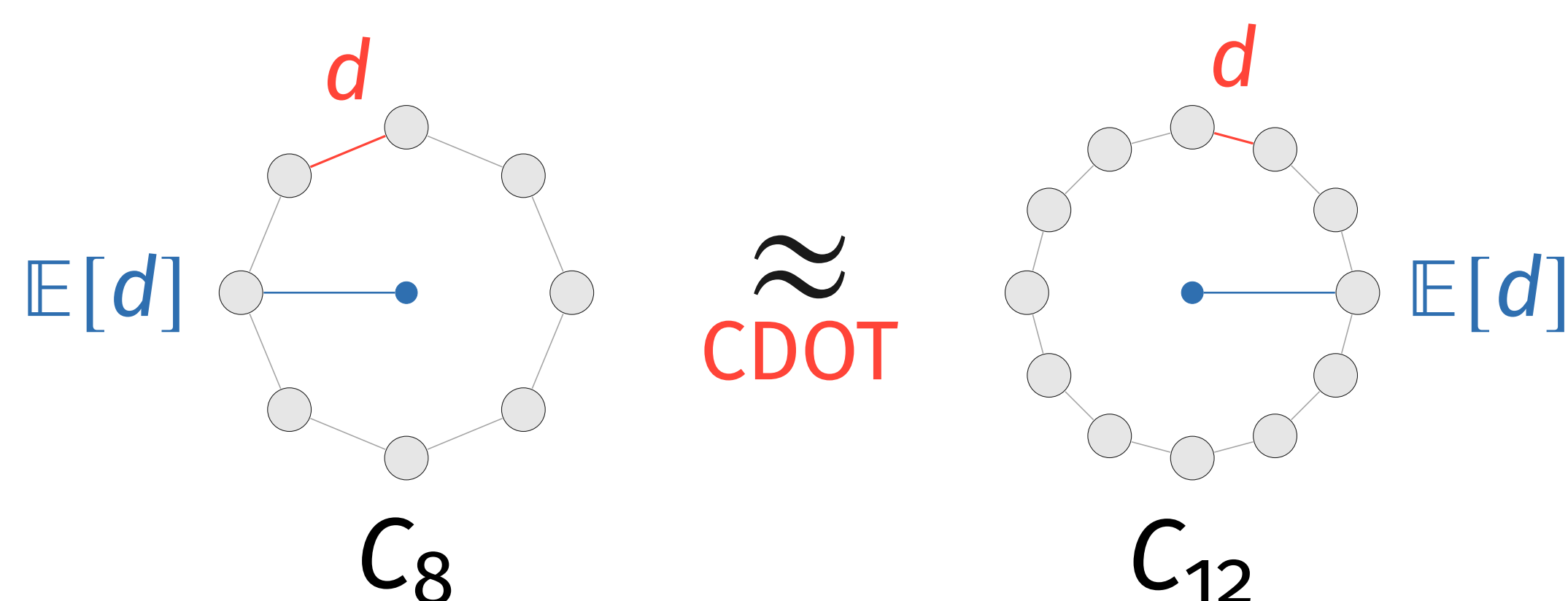
Motivation: comparing whole spaces

- Comparing structured data from different domains (graphs, point clouds, connectomes) needs a **distance between whole spaces**, not between points as in classical optimal transport (OT).
- A canonical approach is to find the coupling that best aligns the two spaces, then takes the leftover mismatch as their distance.
- With a transport plan that aligns the two spaces, such a distance drives graph matching, transfer learning, and shape analysis.
- Gromov-Wasserstein (GW) and its feature-aware FGW do this, but their non-convexity makes theory and optimization hard.

Contributions

- Convexity.** A convex problem with a global optimum.
- Pseudometric.** A valid pseudometric on attributed compact metric-measure (mm) spaces.
- Transport plan.** An explicit plan recovered across domains.
- Statistical guarantees.** A consistent estimator with finite-sample rates, in $O(n^3)$ time (on par with FGW).
- Relation to GW.** GW's non-convexity explained via a dispersion gap.

What CDOT sees: smoothed profiles



- GW matches distances term by term ($|d - d'|$). CDOT instead reads each point through its **aggregated distance profile** $\mathbb{E}[d(x, \cdot)]$ and aligns these profiles across domains.
- The profile is a smoothed, global summary, so C_8 and C_{12} count as the same structure, which is what keeps the objective convex.

Related work

On population mm spaces, no prior method delivers all four at once: a cross-domain plan, a valid metric, a convex objective, and statistical guarantees.

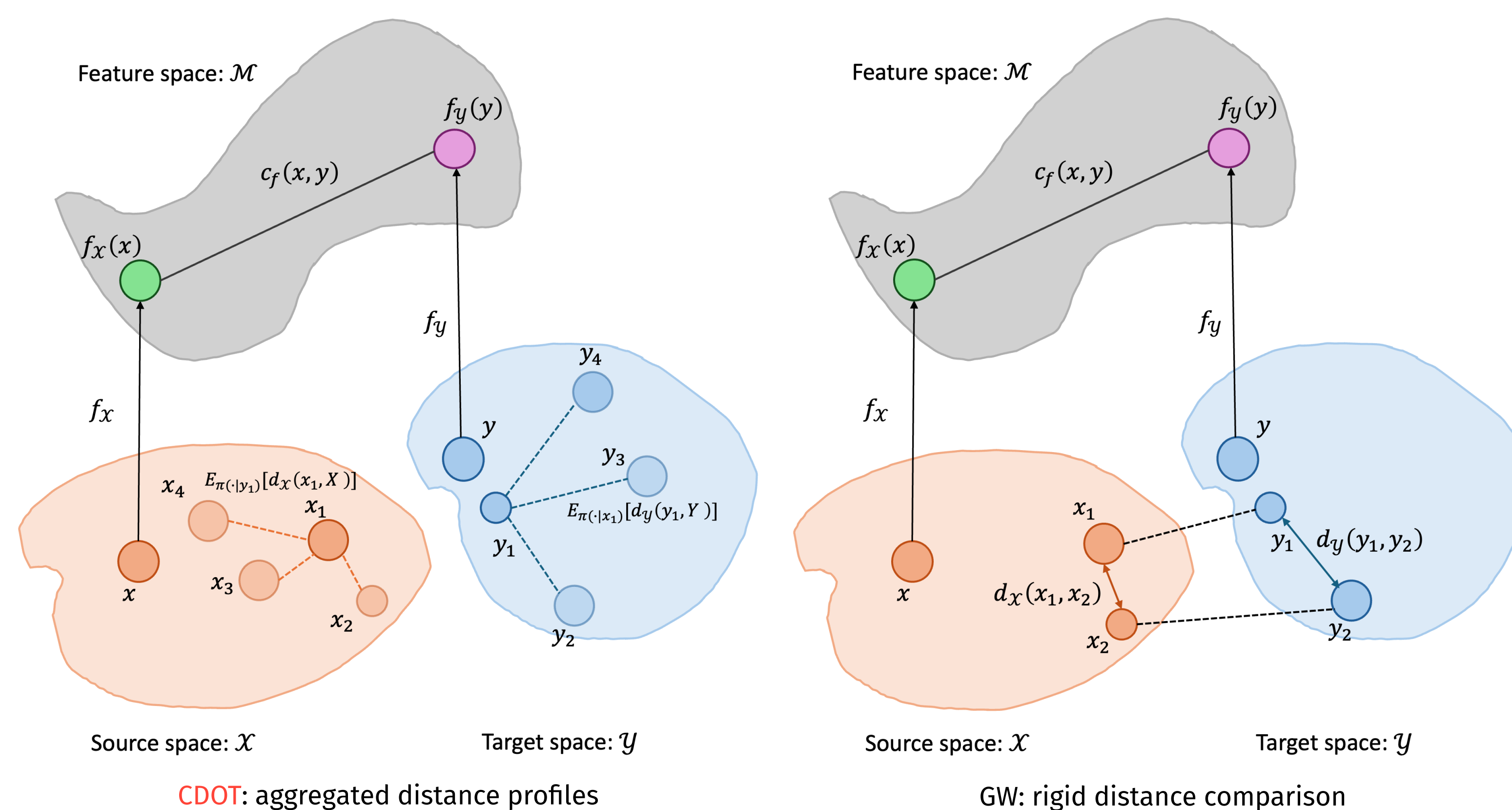
Domain	Method	Plan	Feat.	Time	Theoretical guarantees			
					Metric	Convex	Consist.	Finite
mm spaces	CDOT (Ours)	✓	✓	$O(n^3)$	✓	✓	✓	✓
	GW	✓	✗	$O(n^3)$	✓	✗	△	✗
	FGW	✓	✓	$O(n^3)$	△	✗	△	△
	Entropic GW	✓	✗	$O(n^2)$	✗	△	✗	✗
Euclidean	Sliced GW	✗	✗	$O(n \log n)$	✓	✗	✗	✗
Discrete mm	Low-rank GW	✓	✗	$O(n)$	✗	✗	✗	✗
	GW-SDP	✓	✗	$O(n^6)$	✗	✓	✗	✗
	Sliced FGW	✗	✓	$O(n^2 \log n)$	✓	✗	✗	✗
Graphs	Spectral	△	✗	$O(n^3)$	✗	✗	✗	✗
	IsoRank	✓	✓	$O(n^3)$	✗	✗	✗	✗
	COPT	✓	✗	$O(n^{2.373})$	✓	✗	✗	✗

✓ holds △ partial/weaker ✗ none. Only **CDOT** attains all guarantees while returning a cross-domain plan.

Notations

- $(X, d_X, \mathbb{P}_X)$ and $(Y, d_Y, \mathbb{P}_Y)$ are compact mm spaces, where $\mathbb{P}_X$ and $\mathbb{P}_Y$ are fully supported Borel probability measures.
- $f_X : X \rightarrow \mathcal{M} \subset \mathbb{R}^k$ and $f_Y : Y \rightarrow \mathcal{M}$ are continuous mappings that represent auxiliary feature information, and $\mathcal{M}$ is assumed to be compact.
- $\mathfrak{X} = (X, d_X, \mathbb{P}_X, f_X)$ and $\mathfrak{Y} = (Y, d_Y, \mathbb{P}_Y, f_Y)$ are attributed compact mm spaces.
- For Borel probability measures μ and ν , possibly lying on different spaces, $\Pi(\mu, \nu)$ is the set of couplings whose marginals are μ and ν .

Method: Convex Distance Operator Transport



CDOT objective

$$\mathcal{L}_\alpha(\pi) = (1 - \alpha) \underbrace{\mathbb{E}_\pi[c_f(X, Y)]}_{\text{Feature } \mathcal{F}(\pi)} + \frac{\alpha}{2} \underbrace{\int \int (\mathbb{E}_{\pi(\cdot|y)}[d_X(x, X)] - \mathbb{E}_{\pi(\cdot|x)}[d_Y(y, Y)])^2 \mathbb{P}_X(dx) \mathbb{P}_Y(dy)}_{\text{Structure } \mathcal{R}(\pi)}.$$

Feature information

- The **feature cost** c_f scores feature disagreement across domains: feature maps f_X, f_Y send both to a shared $\mathcal{M}$ (e.g., labels, embeddings), and we set $c_f(x, y) = \|f_X(x) - f_Y(y)\|_2^2$.

Structural information

- The **structural cost** compares **aggregated distance profiles**.
- $\mathcal{R}(\pi) = 0$ when the two spaces' mean-distance profiles align under the coupling.

Theorem (Convexity). For each $\alpha \in [0, 1]$, the CDOT problem $\inf_{\pi \in \Pi(\mathbb{P}_X, \mathbb{P}_Y)} \mathcal{L}_\alpha(\pi)$ is convex and has a global minimizer $\pi^* \in \Pi(\mathbb{P}_X, \mathbb{P}_Y)$.

Pseudometricity

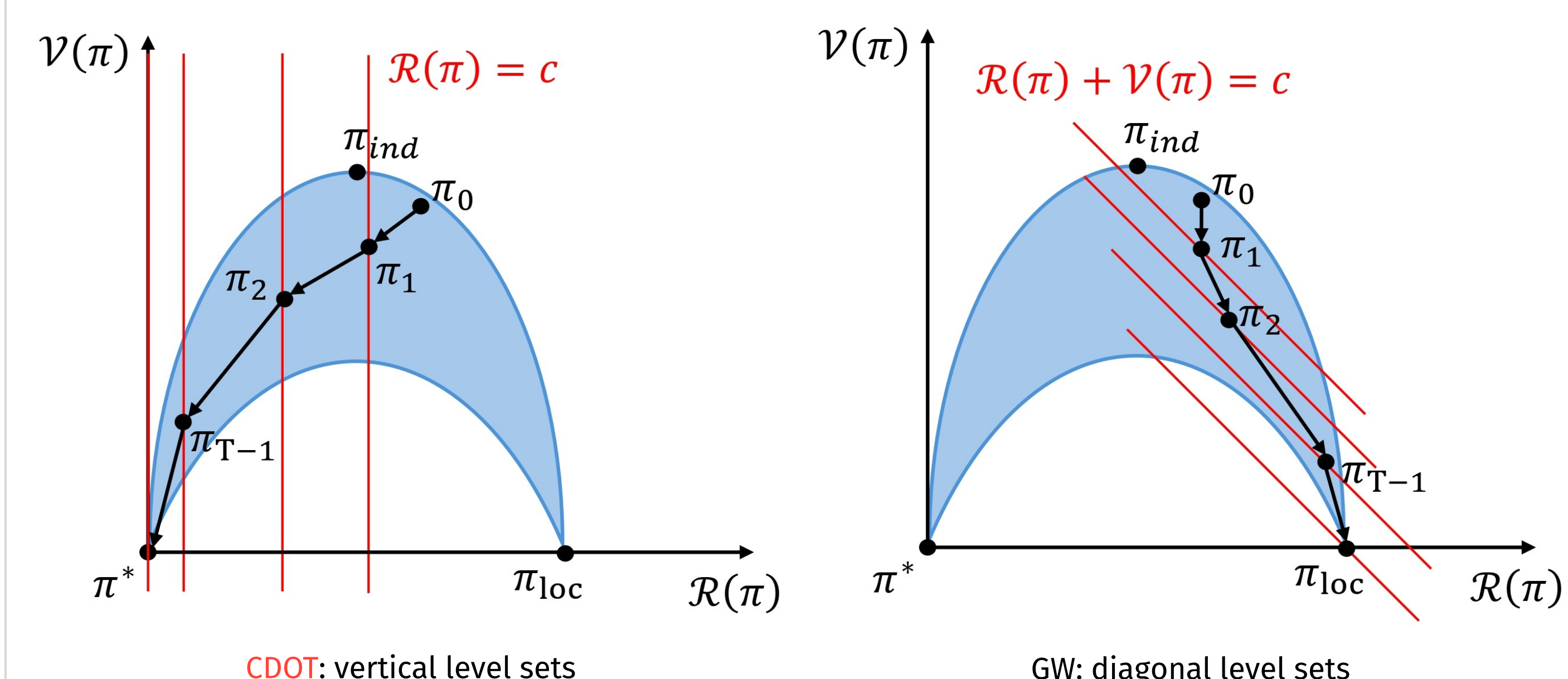
Theorem (Pseudometric). The CDOT discrepancy

$$d_{CT}^{(\alpha)}(\mathfrak{X}, \mathfrak{Y}) = \left(\min_{\pi \in \Pi(\mathbb{P}_X, \mathbb{P}_Y)} \mathcal{L}_\alpha(\pi; \mathfrak{X}, \mathfrak{Y}) \right)^{1/2}$$

is a pseudometric on attributed compact mm spaces.

- Why only "pseudo"?** $d_{CT}^{(\alpha)} = 0$ does not force $\mathfrak{X} = \mathfrak{Y}$; it certifies a coupling that preserves both the feature maps and the distance operators. This **operator-level equivalence** is strictly weaker than the measure-preserving isometry behind GW.
- Concretely** (the C_8 vs C_{12} rings): the two cyclic structures share one aggregated distance profile, so $d_{CT}^{(\alpha)} = 0$, while GW separates them with a positive cost. CDOT identifies structures that GW splits apart.

Relation to GW: the dispersion gap



Conditioning GW's pairwise cost on each coupled pair splits it into a mean part $\mathcal{R}$ and a variance part $\mathcal{V}$: $\mathcal{R}_{GW,2}(\pi) = \mathcal{R}(\pi) + \mathcal{V}(\pi)$, where

$$\mathcal{V}(\pi) = \int \int (\text{Var}_\pi[d_X(x, X) | Y=y] + \text{Var}_\pi[d_Y(y, Y) | X=x]) \mathbb{P}_X(dx) \mathbb{P}_Y(dy).$$

- $\mathcal{V}$ is a **coupling's spread**: it **vanishes** at bi-deterministic plans (a hard matching, where each $\pi(\cdot | x)$ is a point mass) and is **maximal** at the independent coupling $\pi_{\text{ind}} = \mathbb{P}_X \otimes \mathbb{P}_Y$ (the $\mathcal{V}$ -axis of the plots above).
- GW minimizes $\mathcal{R} + \mathcal{V}$, so the concave $\mathcal{V}$ rewards deterministic plans and bends the landscape non-convex. CDOT keeps only $\mathcal{R}$: its vertical level sets reach the global optimum, while GW's diagonal ones can trap it.

Consistency

- The empirical CDOT problem is a **convex QP**; solved by a globally convergent **Frank-Wolfe** (FW) algorithm.
- Theorem (Consistency).** Under some assumptions, the empirical CDOT value **asymptotically achieves** the global optimum, with a non-asymptotic rate decomposed into optimization + statistical error:

$$|\hat{\mathcal{L}}_\alpha(\hat{\pi}) - \mathcal{L}_\alpha(\pi^*)| \leq \underbrace{\frac{32\alpha n}{T+3}}_{\text{Optimization error}} + C \underbrace{\left(W_1^{d_X}(\mathbb{P}_X, \hat{\mathbb{P}}_X) + W_1^{d_Y}(\mathbb{P}_Y, \hat{\mathbb{P}}_Y) \right)}_{\text{Statistical error}} \xrightarrow{n \uparrow \infty} 0,$$

where n is the number of samples, T is the iteration count of FW, and $C > 0$ is a constant.

Empirical validation

Graph classification (TUDataset)

Dataset	Node info.	CDOT (Ours)	FGW	GW	COPT
MUTAG	Label	86.2	82.5	71.8	63.3
IMDB-B	None	64.2	60.2	—	63.7
PROTEINS	Attr.	75.5	73.6	66.1	69.5
NCI1	Label	74.8	73.0	57.1	59.9
ENZYMES	Attr.	51.3	44.5	23.8	23.5

Classification accuracy % (mean).

- CDOT wins on every dataset**, with the biggest gains on feature-rich sets (MUTAG, ENZYMES) and a lead even on feature-less IMDB-B.
- $d_{CT}^{(\alpha)}$ feeds a Gaussian **RBF kernel** $K = \exp(-\gamma D^2)$ into an **SVM** classifier.
- Hyperparameters (e.g., α, γ) are chosen by **nested CV**; $\alpha = 1$ for feature-less IMDB-B.